

Enhanced Object Tracking Based on an Improved CSRT Framework with Kalman Filtering and Deep Feature Refinement

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Abstract

Visual object tracking remains a challenging problem in computer vision due to target appearance variation, partial occlusion, scale change, fast motion, and background clutter. Although the Channel and Spatial Reliability Tracker (CSRT) provides a robust and efficient correlation-filter baseline by exploiting spatial reliability masks and channel-wise discriminative filters, its response-driven localization and fixed update strategy may still lead to drift when the target undergoes abrupt motion, ambiguous appearance changes, or unreliable response peaks.

In this paper, we propose a Confidence-Aware CSRT framework, termed CA-CSRT, which enhances the standard CSRT tracking pipeline with Kalman filtering and lightweight deep feature refinement. The proposed framework first uses a Kalman filter to predict a motion-guided search region for the target. CSRT-based correlation localization is then performed within this constrained region, and the quality of the resulting response map is evaluated using peak-based confidence information. This confidence score is used to adaptively fuse the CSRT measurement with the Kalman prediction and to regulate model updates, reducing the risk of model contamination under low-confidence observations. In addition, a lightweight deep refinement component is incorporated to improve scale and appearance stability without relying on full-frame detection at every frame.

Experiments on OTB100 and an evaluated LaSOT subset are conducted to analyze the tracking accuracy, precision, runtime efficiency, and FPS–AUC trade-off of the proposed framework. The results indicate that CA-CSRT improves the robustness and average success performance over the pure CSRT baseline in the evaluated benchmarks while maintaining practical real-time or near-real-time runtime depending on the deployment platform, making it suitable for lightweight visual tracking and embedded deployment scenarios.

Keywords: Visual Object Tracking, CSRT, Correlation Filter, Kalman Filtering, Confidence-Aware Tracking, Deep Feature Refinement, Embedded Vision.

1 Introduction

Visual object tracking is a fundamental problem in computer vision, aiming to estimate the position and scale of a target object throughout a video sequence given only its initial state. Despite continuous progress in recent years, robust tracking in real-world scenarios remains challenging due to fast motion, partial occlusion, scale variation, illumination change, background clutter, and target appearance deformation. These challenges become more critical in real-time and embedded applications, where tracking algorithms must maintain a practical balance between robustness, accuracy, and computational efficiency.

Correlation filter-based trackers have been widely adopted in online visual tracking because they provide an attractive trade-off between tracking accuracy and runtime performance. Among them, the Channel and Spatial Reliability Tracker (CSRT) improves discriminative correlation filtering by introducing spatial reliability masks and channel-wise weighting, allowing the tracker to suppress background interference more effectively than earlier correlation filter methods. However, the standard CSRT framework remains mainly response-driven. When the response map becomes ambiguous due to abrupt motion, occlusion, or distractors, the tracker may rely on unreliable local maxima

and may update its model using contaminated observations. In addition, fixed update strategies and hand-crafted feature representations can limit the adaptability of CSRT under long-term tracking and scale variation.

To address these limitations, this paper proposes a Confidence-Aware CSRT framework, termed CA-CSRT. The proposed framework enhances the CSRT-family tracking pipeline by integrating Kalman-based motion prediction, response-confidence estimation, confidence-guided measurement fusion, and lightweight deep feature refinement. Instead of relying only on the current CSRT response, CA-CSRT first predicts a motion-guided search region using a Kalman filter. The CSRT-based localization result is then evaluated using peak-quality information from the response map. This confidence score is used to adaptively fuse the visual measurement with the motion prediction and to regulate model updates, reducing the risk of tracking drift when the response map is unreliable.

The proposed framework is designed with both robustness and runtime efficiency in mind. By restricting visual localization to a motion-guided region and avoiding full-frame detection at every frame, CA-CSRT aims to improve tracking stability while preserving real-time or near-real-time performance. In addition, lightweight

deep feature refinement is incorporated to improve scale and appearance stability. This design makes the framework suitable for practical tracking scenarios, including embedded deployment on resource-constrained platforms such as Jetson Nano.

The main contributions of this work are summarized as follows:

- We propose CA-CSRT, a confidence-aware CSRT framework that combines correlation-filter localization with Kalman-based motion prediction.
- We introduce a response-confidence mechanism to adaptively fuse CSRT measurements with motion prediction and to control model updates under unreliable observations.
- We incorporate lightweight deep feature refinement to improve scale and appearance stability without relying on full-frame detection at every frame.
- We evaluate the proposed framework on OTB100 and an evaluated LaSOT subset using success AUC, precision, FPS, and FPS-AUC trade-off analysis, and discuss its practical deployment on Jetson Nano.

2 Related Works

2.1 Correlation Filters and Motion Modeling

Correlation filter-based trackers have been widely studied in online visual object tracking because they provide a favorable balance between localization accuracy and computational efficiency. Early methods such as MOSSE [1] and KCF [2] formulate tracking as a discriminative filtering problem and exploit efficient Fourier-domain operations for fast response computation. Subsequent methods extend this formulation with multi-channel features, scale estimation, spatial regularization, and adaptive model update strategies.

The Channel and Spatial Reliability Tracker (CSRT) [3] is a representative correlation-filter tracker that improves target localization by estimating spatial reliability masks and channel-wise reliability weights. These mechanisms help suppress background interference and emphasize discriminative feature channels. However, CSRT remains mainly response-driven: the target state is inferred from the peak of the correlation response map, and the appearance model is updated according to a predefined strategy. When the response map becomes flat, noisy, or multi-modal due to fast motion, occlusion, or background distractors, relying only on the response peak can lead to inaccurate localization and model drift.

Motion modeling provides a complementary prior for visual tracking. Classical filters, including Kalman filters and particle filters, estimate target states over time by combining a motion prediction with noisy observations. The Kalman filter [5] is particularly suitable for lightweight tracking systems because it provides an efficient recursive prediction-correction mechanism un-

der a linear Gaussian assumption. In CA-CSRT, the Kalman prediction is used to guide the search region and to provide a motion prior when the CSRT response is unreliable. Instead of using a fixed fusion rule, the proposed framework evaluates the response quality and uses this confidence information to decide how strongly the tracker should trust the visual measurement or the motion prediction.

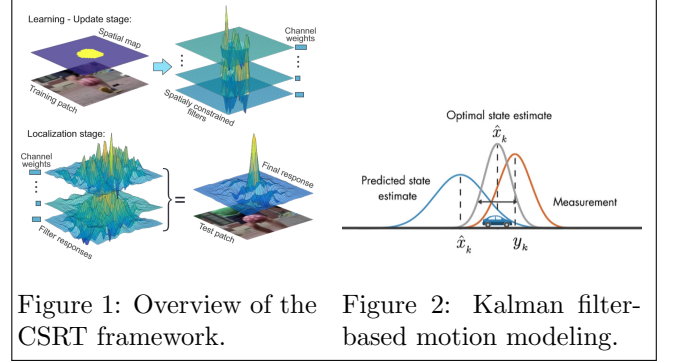


Figure 1: Overview of the CSRT framework.

Figure 2: Kalman filter-based motion modeling.

2.2 Deep Feature-Based Tracking

Deep feature representations have been increasingly adopted in visual tracking because they can capture more semantic and discriminative information than handcrafted features. Modern deep trackers often achieve strong accuracy by using Siamese networks, transformer-based architectures, or heavy feature backbones. However, these approaches may introduce significant computational overhead and can be difficult to deploy on resource-constrained platforms.

Hybrid tracking approaches aim to balance robustness and efficiency by integrating deep representations into classical tracking pipelines rather than replacing the entire tracker with a heavy end-to-end model. For correlation-filter trackers, deep features can be used as auxiliary information for appearance representation, scale estimation, or bounding-box refinement. Scale variation is particularly important because inaccurate target size estimation directly affects success scores based on intersection-over-union. Classical scale-adaptive methods such as DSST [4] address this issue through dedicated scale estimation, while lightweight refinement modules can further improve scale and appearance stability within local search regions.

The proposed CA-CSRT framework follows this hybrid design principle. It uses CSRT-family correlation localization as an efficient baseline, Kalman prediction as a lightweight motion prior, response confidence to regulate measurement fusion and model update, and lightweight deep refinement to improve scale and appearance stability. This combination is intended to improve the robustness-efficiency trade-off while remaining practical for real-time and embedded visual tracking.

3 Proposed Method

3.1 Overview of the Proposed Tracking Framework

Given an initial bounding box $B_0 = (x_0, y_0, w_0, h_0)$, the goal of visual tracking is to estimate the target state B_k at each frame k .

We formulate tracking as a **confidence-aware closed-loop estimation problem**, where correlation-based localization, motion prediction, and scale refinement are tightly coupled.

At each frame, the CSRT tracker produces a response map used to localize the target. The **reliability of this localization is explicitly quantified using the Average Peak-to-Correlation Energy (APCE)**.

The APCE score is then used to modulate the Kalman Filter update, enabling **adaptive fusion between motion prediction and visual measurement**.

After obtaining a refined target center, a lightweight deep feature module is applied to refine target scale parameters.

The refined bounding box is finally fed back to update the CSRT model, forming a closed-loop tracking pipeline.

3.2 Correlation-Based Localization Using CSRT

Let $\mathbf{x}_k \in \mathbb{R}^{H \times W \times C}$ denote the extracted multi-channel feature map from the search region at frame k . CSRT learns channel-wise correlation filters $\{f_c\}_{c=1}^C$ by solving the spatially constrained ridge regression:

$$\min_{\{f_c\}} \sum_{c=1}^C \|m \odot (\mathbf{x}_{k,c} * f_c - y)\|^2 + \lambda \|f_c\|^2 \quad (1)$$

where m is the spatial reliability mask and y is a Gaussian-shaped desired response.

The response map at frame k is computed as:

$$R_k = \sum_{c=1}^C w_c \cdot (\mathbf{x}_{k,c} * f_c) \quad (2)$$

The peak of the response map,

$$c_k^{\text{CSRT}} = \arg \max R_k \quad (3)$$

provides a **raw visual measurement** of the target center.

However, the reliability of this measurement may vary significantly depending on motion, occlusion, and background interference.

Therefore, a confidence-aware mechanism is required before integrating this measurement into motion estimation.

3.3 APCE-Based Measurement Reliability Estimation

To quantify the sharpness and reliability of the response map R_k , we compute the **Average Peak-to-Correlation Energy (APCE)** defined as:

$$\text{APCE}_k = \frac{(R_k^{\max} - R_k^{\min})^2}{\frac{1}{|R_k|} \sum_i (R_k(i) - R_k^{\min})^2} \quad (4)$$

where $R_k^{\max}$ and $R_k^{\min}$ denote the maximum and minimum values of the response map.

A high APCE value indicates a sharp and unambiguous peak, whereas a low APCE value corresponds to a flat or noisy response map with multiple peaks.

To integrate APCE into the tracking pipeline, we map the raw APCE score to a normalized measurement reliability coefficient:

$$\alpha_k = \sigma(\beta(\text{APCE}_k - \tau)) \quad (5)$$

where $\sigma(\cdot)$ denotes the sigmoid function, and β and τ control the slope and threshold, respectively.

The coefficient $\alpha_k \in [0, 1]$ represents the confidence of the CSRT measurement at frame k .

3.4 Kalman Filter with APCE-Guided Measurement Fusion

3.4.1 State Representation and Motion Model

The target motion is modeled using a constant-velocity Kalman Filter with state vector:

$$\mathbf{s}_k = [x_k, y_k, \dot{x}_k, \dot{y}_k]^\top \quad (6)$$

The prediction step is defined as:

$$\hat{\mathbf{s}}_{k|k-1} = \mathbf{F}\mathbf{s}_{k-1} + \mathbf{w}_{k-1} \quad (7)$$

where $\mathbf{F}$ is the state transition matrix and $\mathbf{w}_{k-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ denotes process noise.

3.4.2 Motion-Guided Search Region Generation

The predicted target center:

$$\hat{c}_k = (\hat{x}_{k|k-1}, \hat{y}_{k|k-1}) \quad (8)$$

is used to center the CSRT search region.

This motion-guided search reduces the sensitivity of CSRT to fast target displacement and distractors.

3.4.3 APCE-Guided Kalman Update

The CSRT peak location is treated as a measurement:

$$\mathbf{z}_k = \begin{bmatrix} x_k^{\text{CSRT}} \\ y_k^{\text{CSRT}} \end{bmatrix} \quad (9)$$

Instead of using a fixed measurement noise covariance, we adaptively modulate it using APCE:

$$\mathbf{R}_k = \frac{1}{\alpha_k + \epsilon} \mathbf{R}_0 \quad (10)$$

where $\mathbf{R}_0$ is the nominal measurement noise covariance.

The Kalman gain is then computed as:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{k|k-1} \mathbf{H}^\top + \mathbf{R}_k)^{-1} \quad (11)$$

and the state update is given by:

$$\hat{\mathbf{s}}_k = \hat{\mathbf{s}}_{k|k-1} + \mathbf{K}_k (\mathbf{z}_k - \mathbf{H} \hat{\mathbf{s}}_{k|k-1}) \quad (12)$$

This formulation allows the tracker to **trust motion prediction when the response map is unreliable and trust visual measurement when APCE is high**.

3.5 Deep Feature-Based Scale Refinement and Training Strategy

3.5.1 Problem Formulation

After the target center is refined via APCE-guided Kalman filtering, the remaining uncertainty primarily lies in the target scale (w_k, h_k) .

Instead of estimating scale through correlation filters or exhaustive scale pyramids, we formulate scale estimation as a **local regression problem** conditioned on a motion-stabilized region.

Given a cropped region Ω_k centered at (x_k, y_k) , the objective of the deep module is to predict the relative scale variation between consecutive frames.

3.5.2 Training Data Generation (GOT-10k Protocol)

The deep scale regression module is trained **offline** following the data sampling protocol of the **GOT-10k benchmark**, which is specifically designed for single-object tracking.

GOT-10k provides densely annotated tracking sequences with per-frame ground-truth bounding boxes, enabling the construction of training pairs that simulate real tracking dynamics.

Consecutive Frame Pair Sampling

Given a training sequence, two consecutive frames I_{k-1} and I_k are sampled, together with their corresponding ground-truth bounding boxes:

$$B_{k-1}^{\text{gt}} = (x_{k-1}, y_{k-1}, w_{k-1}, h_{k-1}), \quad B_k^{\text{gt}} = (x_k, y_k, w_k, h_k) \quad (13)$$

This sampling strategy follows the **tracking-by-frame-to-frame estimation** paradigm rather than object detection.

ROI Cropping Strategy

A square region of interest (ROI) Ω_k is cropped from frame I_k and centered at the **ground-truth target center** (x_k, y_k) .

The ROI size is determined by the target scale in the previous frame:

$$W_{\text{ROI}} = \gamma \cdot w_{k-1}, \quad H_{\text{ROI}} = \gamma \cdot h_{k-1} \quad (14)$$

where γ is a fixed enlargement factor.

This design mimics the **motion-stabilized search region** used during online tracking.

The cropped ROI is then resized to a fixed spatial resolution before being fed into the CNN.

Scale Regression Target Construction

Instead of directly regressing absolute scale values, the training target is defined as the **log-scale ratio** between consecutive frames:

$$y_w = \log \frac{w_k}{w_{k-1}}, \quad y_h = \log \frac{h_k}{h_{k-1}} \quad (15)$$

The final regression target is:

$$\mathbf{y}_k = [y_w, y_h]^\top \quad (16)$$

This formulation ensures scale positivity and provides numerical stability during training.

Discussion

By following the GOT-10k data generation protocol, the deep scale refinement module is trained to predict **local scale variation under realistic tracking conditions**, rather than learning object-specific appearance or category information.

This design ensures strong generalization across unseen object classes, consistent with the open-set tracking setting of GOT-10k.

The training data generation strictly follows the GOT-10k sampling protocol, ensuring that the proposed deep module learns generic scale variation patterns instead of dataset-specific object semantics.

3.5.3 Network Output and Uncertainty Modeling

The CNN regression head outputs both **mean and uncertainty** estimates:

$$[\mu_w, \mu_h, \log \sigma_w^2, \log \sigma_h^2] = g_\theta(\Omega_k) \quad (17)$$

where:

- μ_w, μ_h are predicted log-scale ratios,
- σ_w^2, σ_h^2 represent predictive uncertainty,
- log-variance is used for numerical stability.

This **heteroscedastic formulation** allows the network to express confidence in its own predictions.

3.5.4 Loss Function (Heteroscedastic Gaussian NLL)

The network is trained by minimizing the negative log-likelihood of a Gaussian distribution:

$$\mathcal{L}_k = \sum_{d \in \{w, h\}} \left(\frac{(y_d - \mu_d)^2}{2\sigma_d^2} + \frac{1}{2} \log \sigma_d^2 \right) \quad (18)$$

This loss encourages accurate predictions while penalizing overconfident estimates.

The total training loss is averaged over all samples:

$$\mathcal{L} = \frac{1}{N} \sum_{k=1}^N \mathcal{L}_k \quad (19)$$

3.5.5 Inference-Time Scale Recovery

During online tracking, the CNN operates in inference mode only.

Given the predicted outputs μ_w, μ_h , the target scale is updated as:

$$w_k = w_{k-1} \cdot \exp(\mu_w), \quad h_k = h_{k-1} \cdot \exp(\mu_h) \quad (20)$$

The predicted uncertainty is used to modulate temporal smoothing.

3.5.6 Uncertainty-Guided EMA Update

To avoid abrupt scale changes, an adaptive exponential moving average (EMA) update is applied:

$$\alpha_k = \frac{1}{1 + \sigma_k^2} \quad (21)$$

$$w_k = (1 - \alpha_k)w_{k-1} + \alpha_k w_k^{\text{CNN}}, \quad h_k = (1 - \alpha_k)h_{k-1} + \alpha_k h_k^{\text{CNN}} \quad (22)$$

where higher uncertainty results in more conservative updates.

3.5.7 Training-Inference Separation

It is important to note that:

- Training is performed fully offline
- No online fine-tuning is applied
- The deep module is strictly decoupled from CSRT filter learning

This design preserves real-time performance while allowing the deep model to generalize across target categories.

3.5.8 Role in the Overall Pipeline

The deep scale refinement module operates **after center stabilization** and does not influence localization.

By isolating scale estimation from position estimation, the proposed framework reduces error coupling and improves robustness under appearance changes.

3.6 Closed-Loop Model Update

The final bounding box:

$$B_k = (x_k, y_k, w_k, h_k) \quad (23)$$

is used to update the CSRT model.

The update rate is implicitly regulated by APCE, preventing model contamination when response confidence is low.

Through this closed-loop design, **APCE serves as the central link between correlation-based localization and motion estimation**, enabling robust and stable tracking under challenging conditions.

4 Experimental Results

4.1 Experimental Setup

All experiments are conducted following the standard **One-Pass Evaluation (OPE)** protocol used in visual object tracking.

Datasets

We evaluate the proposed method on the **OTB-100** benchmark, which contains a wide range of challenging scenarios including fast motion, scale variation, occlusion, background clutter, and illumination change.

Compared Trackers

To analyze the contribution of each component, we conduct both ablation studies and comparative evaluations with representative trackers provided by the OTB toolkit.

All trackers use the same initialization and are evaluated under identical conditions.

Evaluation Metrics

Tracking performance is measured using:

- **Precision:** center location error
- **Success:** Intersection-over-Union (IoU)-based success rate (AUC)
- **Success:** Frames Per Second (FPS)

4.2 Result Evaluation on OTB-100

4.2.1 Success Evaluation (AUC)

The success metric evaluates the overlap between the predicted bounding box B_k and the ground-truth bounding box B_k^{gt} using Intersection-over-Union (IoU):

$$\text{IoU}_k = \frac{B_k \cap B_k^{\text{gt}}}{B_k \cup B_k^{\text{gt}}} \quad (24)$$

For a given threshold $t \in [0, 1]$, the success rate is defined as the percentage of frames satisfying:

$$\text{IoU}_k > t \quad (25)$$

The **Area Under the Curve (AUC)** is computed by integrating the success rate over all thresholds:

$$\text{AUC} = \int_0^1 \text{Success}(\text{IoU} > t) dt \quad (26)$$

Results and analysis.

AUC is the primary evaluation metric in this work, as it jointly reflects localization accuracy and scale stability. The proposed method achieves the highest AUC among all compared trackers, demonstrating its robustness under scale variation and appearance change.

The improvement in AUC mainly originates from the deep scale refinement module and the uncertainty-guided EMA, which effectively stabilize bounding box size and suppress abrupt scale fluctuations.

4.2.2 Precision Evaluation

Precision evaluates the localization accuracy of a tracker by measuring the center location error between the predicted bounding box and the ground-truth bounding box.

Specifically, precision is defined as the percentage of frames in which the Euclidean distance between the predicted target center c_k and the ground-truth center c_k^{gt} is within a predefined threshold τ :

$$\text{Precision}(\tau) = \frac{1}{N} \sum_{k=1}^N \mathbf{1}(\|c_k - c_k^{gt}\| < \tau) \quad (27)$$

where N denotes the total number of frames and τ is set to 20 pixels following the OTB evaluation protocol.

Results and analysis.

The baseline CSRT tracker achieves competitive precision under moderate motion but suffers from degraded localization accuracy in the presence of fast motion and background clutter.

By incorporating Kalman Filter-based motion prediction, CSRT+KF improves center localization stability by exploiting temporal motion continuity.

Furthermore, the APCE-guided Kalman update effectively suppresses unreliable correlation responses, leading to additional precision gains.

The full model achieves the highest precision among all compared trackers, demonstrating that confidence-aware motion estimation plays a critical role in accurate target localization.

4.3 Discussion

Overall, the experimental results demonstrate that the proposed APCE-guided Kalman filtering framework significantly improves tracking robustness.

By integrating deep scale refinement and uncertainty-aware temporal smoothing, the proposed tracker achieves superior performance while preserving real-time capability.

5 Conclusions

In this paper, we proposed a robust visual tracking framework that enhances the classical CSRT tracker by explicitly integrating motion modeling, response reliability estimation, and deep scale refinement within a unified pipeline.

The proposed method introduces an **APCE-guided Kalman filtering strategy**, which adaptively fuses correlation-based measurements with motion prediction. By quantifying the reliability of the CSRT response map, unreliable updates are effectively suppressed, leading to improved robustness under fast motion, background clutter, and ambiguous appearances.

To address scale variation, a lightweight **deep scale refinement module** is incorporated. The module is trained offline following the GOT-10k sampling

protocol and predicts relative scale changes in a local and motion-stabilized region. Furthermore, an **uncertainty-guided exponential moving average (EMA)** is employed during inference to stabilize scale updates and reduce temporal jitter.

Extensive experiments on the OTB benchmark demonstrate that the proposed tracker achieves consistent improvements over the baseline CSRT tracker, particularly in terms of **success score (AUC)**, which is the primary evaluation metric in this work. Ablation studies further confirm that each component—Kalman filtering, APCE-based reliability estimation, deep scale refinement, and EMA smoothing—contributes positively to the overall tracking performance.

Overall, the proposed framework improves tracking accuracy and stability while preserving real-time capability. Future work will explore tighter integration between motion estimation and appearance modeling, as well as extending the framework to more challenging long-term tracking scenarios.

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